ModelIt!

Student Activity Pack

Can Renewable Energy Replace Fossil Fuels?

Modeling Energy Systems, Grid Infrastructure, and the Clean Energy Transition

Grade 12L2 | NGSS HS-ESS3-2 | Energy and Matter

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Where does the electricity in your home come from? Can you trace it from the power plant to your outlet?**

2. **What happens when the sun goes down or the wind stops blowing if a region depends on solar and wind power? How could this problem be solved?**

3. **If renewable energy is now cheaper than fossil fuels, why do fossil fuels still provide 80% of global energy? What barriers prevent the switch?**

4. **What do you think a 100% renewable energy grid would require beyond just solar panels and wind turbines?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Total Energy Demand | Renewable Generation Capacity | Fossil Fuel Generation | Energy Storage Capacity | Grid Reliability Index | Carbon Emission Rate | Energy Cost Per Kilowatt-Hour

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Renewables Without Storage

Renewable Capacity: Doubled | Storage: Current | Grid: Current Infrastructure

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Storage Breakthrough

Renewable Capacity: Current Growth | Storage: Tripled | Grid: Current Infrastructure

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Full System Transition

Renewable Capacity: Doubled | Storage: Tripled | Grid: Modernized | Demand: Flexible

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Nuclear Baseload Generation — Electricity from nuclear fission that provides constant, carbon-free baseload power with 90%+ capacity factor — nuclear complements intermittent renewables by providing reliable generation but faces challenges of high construction costs, long build times, waste storage, and public opposition

Electric Vehicle Grid Integration — The use of electric vehicle batteries as distributed energy storage for the grid — millions of EVs plugged in during peak solar hours could absorb excess generation, and vehicle-to-grid technology could discharge stored energy during peak demand, turning the transportation fleet into a massive battery

Hydrogen Energy Storage — The production of hydrogen through electrolysis using excess renewable electricity — hydrogen can be stored for weeks to months and converted back to electricity or used directly as fuel, addressing the seasonal storage challenge that batteries cannot solve cost-effectively

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Renewable energy is now cheaper than fossil fuels in most of the world, yet fossil fuels still provide 80% of global energy — so what technical, economic, and political barriers prevent us from making a transition that is already economically rational? Use evidence from your simulation to support your answers.

1. **Why is adding more renewable capacity insufficient without proportional investment in storage and grid infrastructure? What does your model reveal about the intermittency challenge?**

2. **Your model showed that storage is the keystone technology. What types of storage are needed for different timescales — hourly, daily, and seasonal — and what technologies address each?**

3. **How does the concept of capacity factor explain why a 100% renewable grid needs much more installed generation capacity than a fossil fuel grid serving the same demand?**

4. **What role should nuclear energy play in the clean energy transition? Evaluate the trade-offs using evidence from your model about baseload reliability and carbon reduction.**

5. **The technology for 100% renewable energy exists today. What non-technical barriers — political, economic, social — are preventing faster deployment, and how would you address them?**

8 STEM Challenge: Design a 100% Renewable Energy Grid

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YOUR ENGINEERING MISSION

Design a regional energy system that achieves 100% renewable electricity generation while maintaining grid reliability above 99.97%, using your model evidence to determine the optimal mix of generation, storage, and demand management.

THE SCENARIO

A state with 5 million people currently generates 60% of its electricity from natural gas, 20% from coal, 15% from solar and wind, and 5% from hydroelectric. The governor has committed to 100% clean electricity by 2045. Your engineering team must design the generation, storage, and grid infrastructure plan that achieves this goal while maintaining reliable power delivery through all weather conditions and demand peaks. Your model data on intermittency, storage requirements, and grid stability should guide every design decision.

Engineering Design Thinking

1. WHAT SPECIFIC MIX OF SOLAR, WIND, AND OTHER RENEWABLE GENERATION DOES YOUR PLAN INCLUDE, AND WHAT MODEL EVIDENCE SUPPORTS THIS MIX?

2. HOW MUCH ENERGY STORAGE DOES YOUR PLAN REQUIRE, AND WHAT STORAGE TECHNOLOGIES DOES IT USE FOR DIFFERENT TIMESCALES (HOURS, DAYS, SEASONS)?

3. WHAT GRID INFRASTRUCTURE UPGRADES DOES YOUR PLAN REQUIRE, INCLUDING TRANSMISSION LINES, SMART GRID TECHNOLOGY, AND INTERCONNECTION WITH NEIGHBORING REGIONS?

4. HOW DOES YOUR PLAN ENSURE ENERGY AFFORDABILITY AND EQUITABLE ACCESS THROUGHOUT THE TRANSITION, ESPECIALLY FOR LOW-INCOME HOUSEHOLDS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Renewable energy engineers who design solar farms, wind installations, and energy storage systems earn $70,000-$140,000/year. Grid systems engineers who manage the complex task of balancing variable renewable generation with reliable power delivery earn $80,000-$150,000/year at utilities and grid operators. Energy policy analysts who design regulatory frameworks, carbon pricing mechanisms, and clean energy incentives earn $65,000-$130,000/year in government agencies and think tanks.